

Extreme weather

Tornadoes, hurricanes, and flash floods destroy homes and countryside. Long dry spells cause water shortages and parched crops. There have always been episodes of extreme weather, but now it seems likely that the increase in freak events all around the world is due to climate change.

CAUSES OF EXTREME WEATHER

The Sun's heat is a key factor, creating excessively high and low atmospheric pressure that can lead to extreme weather conditions. Dust from volcanoes can cause major disturbance, and global warming may play a part.



GLOBAL WARMING
Since 1970 global temperatures have risen by 0.5°C (0.9°F), adding heat that may alter weather patterns.

WHAT IS EXTREME WEATHER?

Many parts of the world experience wide variations in their weather, so when does it become extreme? In India torrential monsoons are normal, as is a big freeze in the far North. Put simply, extreme weather is weather that is windier, hotter, colder, wetter, or more destructive than usual.

THE USA HAS THE MOST TORNADOES IN THE WORLD - ABOUT 1,000 EVERY YEAR



THUNDERSTORM

Thunderclouds form in hot, humid weather and bring heavy rain, hail, lightning, and thunder.



MONSOON

These torrential rains that last for weeks are just seasonal weather in subtropical regions.



FLOOD

Too much rain in a short time may cause flash floods in valleys and near rivers and the sea.



TORNADO

These twisting columns of wind can flatten houses and pick up vehicles.



HURRICANE

The Earth's most powerful weather systems bring huge winds and rain.



DUST STORM

In very dry places, sand and soil is picked up in the wind that grows into a suffocating dust storm.



HEATWAVE

During a heatwave, temperatures soar, reservoirs dry up, and water is in short supply.



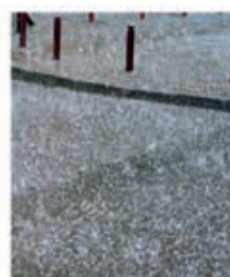
DROUGHT

If there is high pressure for long periods, no clouds form and there is no rain. Vital crops fail.



SEVERE FOG

The thickest fogs occur in polluted areas. Tiny droplets of water settle on particles in the air.



HAILSTORM

Showers of large hailstones can break glass and leave drifts of ice.



SNOWSTORM

A snowstorm is a rapid fall of snow, 15 cm (6 in) deep or more, that disrupts daily life.



COLD WAVE

This dramatic dip in temperature to well below freezing can threaten lives.

MONSOON

Massive monsoon winds bring torrential rain to subtropical regions in summer. This rain is essential for crops to grow. The winds change direction in winter to bring dry, cooler weather.



SUMMER

The South Asian monsoon blows from the Indian Ocean, bringing rain across India.

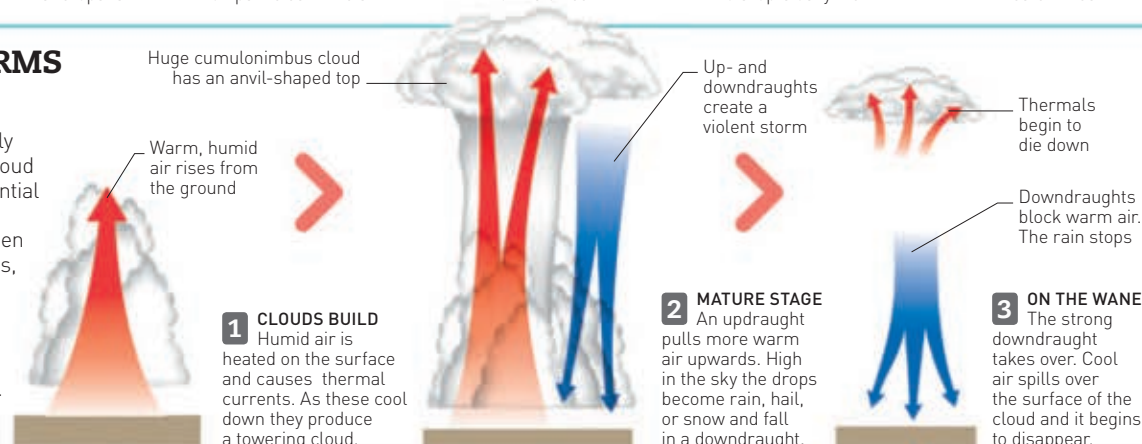


WINTER

Fine, dry weather spreads across India when the South Asian monsoon reverses.

THUNDERSTORMS

In hot, humid weather, an enormous cloud called a cumulonimbus can rapidly build up. This towering cloud brings gusty winds, torrential rain, hail, and lightning. Flashes of lightning happen after droplets, ice crystals, and hail in the cloud become electrically charged. The flashes superheat the air, creating claps of thunder.



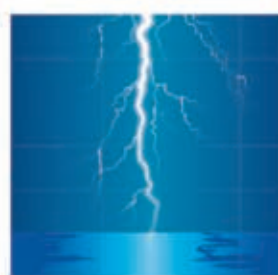
LIGHTNING STROKES

It takes a huge voltage of electricity to overcome the resistance of the air, but once the process starts, strokes of lightning zig-zag towards the ground. When a leader stroke makes contact with a high point like a tree, it lights up with a brighter stroke called the return stroke.



CLOUD-TO-CLOUD LIGHTNING

This most common type of lightning flashes from cloud to cloud then disappears in the air.



CLOUD TO GROUND

Electricity in the lightning joins currents rising from the ground.



RIBBON LIGHTNING

Return strokes flowing back up the first stroke create a ribbon effect.



SHEET LIGHTNING

Lightning flashing inside a cloud looks like a sheet of light.